

Pandemic versus pandemonium: fighting on two fronts

The ongoing COVID-19 pandemic has motivated many across the globe to adopt different approaches to understand the virus, surveil how the virus spreads, and search for effective treatments. This health crisis has focused global efforts on leveraging technology to fight the pandemic. Technological tools have undeniable strengths and hold great promise; however, under these extreme circumstances, where rapid results are expected, there are several challenges to be overcome.

One of the most discussed technological tools has been the development of contact-tracing apps. Many apps now exist to both digitally manage the notification procedure and to track the geographical spread of the virus. These apps will continue to be important as global lockdowns are eased. However, their success is dependent on uptake, and concerns around personal data privacy might impede their use. A survey of 4917 adults in the USA suggests that the population is divided on whether it is acceptable for the government to use mobile phone data to track their location for the purposes of understanding the spread of the virus, for contact tracing, and for monitoring adherence to policies of social distancing. In a move echoing public cries for data security and transparency, the UK's National Health Service contact-tracing app is being overseen by an ethics advisory board, and the software code has been made publicly available. Likewise, the Australian Government is drafting legislation for its COVIDSafe app to protect data privacy.

Candidate treatments are also being augmented by technological tools. Artificial intelligence has been used by companies such as Exscientia, Scipher Medicine, and BenevolentAI to screen for drugs that can be repurposed against the virus. BenevolentAI identified baricitinib, an anti-inflammatory drug used for the treatment of rheumatoid arthritis that is predicted to have antiviral properties. A clinical trial investigating a combined treatment regimen of baricitinib and the antiviral remdesivir began in May; however, given that the drug's safety warnings include a risk of developing serious infections and viral reactivation, both the manufacturer and researchers have cautioned clinicians and patients against the use of baricitinib for COVID-19 treatment until evidence of the efficacy and safety of doing so has been established. Such caution is warranted, given communication of a potential treatment can

lead to self-medication triggered by the spread of misinformation online, as was the case with chloroquine and hydroxychloroquine.

Communication itself has changed dramatically during the pandemic. Global lockdowns have seen a surge in WiFi and mobile phone usage, with social media being one of the major activities. Social media platforms have served as mechanisms to help people stay connected, deliver important public health messages, and disseminate important research findings. However, these platforms have also given rise to the so-called infodemic (an overload of information about the pandemic), making it difficult to distinguish what information is accurate and what is not. Although tech giants such as Facebook and Twitter are attempting to counteract this danger by directing users to official information sources and removing posts that conflict with public health policies, these moves have not quite stemmed the infodemic, suggesting tighter regulations are required. A poll of 1006 adults in the UK suggests that 33% want social media sites to take voluntary action to prevent the spread of misinformation, whereas 55% would be in favour of the UK Government imposing compulsory action, thus directly holding these sites accountable for what is shared on their platforms. Ensuring that such actions are regulated in a fair and transparent manner with the public's best interests at heart will be key.

A general theme here is the effect these technological tools have had on the general public. On the one hand, knowledge is power: being aware of the fast pace at which researchers and companies are developing ways to track COVID-19, treat it, and disseminate information can go some way to making people feel safe. On the other hand, the influence this knowledge can have on them, such as feeling as though their personal data are being monitored, self-medicating with a drug suspected to be effective against the virus, or sharing unsubstantiated claims about the virus on social media, could be dangerous and lead to public mistrust. Ensuring data privacy, appropriately tempering premature findings, and regulating what health information is shared online are necessary steps to overcome these challenges. ■ *The Lancet Digital Health*

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For more on the **Pew Research Center** survey see <https://www.pewresearch.org/fact-tank/2020/04/16/most-americans-dont-think-cellphone-tracking-will-help-limit-covid-19-are-divided-on-whether-its-acceptable/>

For more on the **UK's contact-tracing app** see <https://www.nhs.uk/blogs/digital-contact-tracing-protecting-nhs-and-saving-lives/>

For more on the **COVIDSafe app draft legislation** see <https://www.ag.gov.au/RightsAndProtections/Privacy/Pages/COVIDSafelegislation.aspx>

For more on the **BenevolentAI identification of baricitinib** see [Correspondence Lancet 395: e30-31](https://www.thelancet.com/Correspondence/2020/04/16/395:e30-31)

For more on **cautions regarding baricitinib for COVID-19** see <https://www.nytimes.com/2020/04/13/health/coronavirus-arthritis-drug-baricitinib.html>

For more on **Facebook's response to COVID-19** see <https://about.fb.com/news/2020/04/covid-19-misinfo-update/>

For more on **Twitter's response to COVID-19** see https://blog.twitter.com/en_us/topics/company/2020/covid-19.html

For more on the **Open Knowledge Foundation poll** see <https://okfn.org/about/press/releases/opinion-poll-majority-brits-want-government-action-against-online-disinformation/>